

Securely exposing web services on Open OnDemand

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Abstract

Our multidisciplinary team of medical researchers, software engineers, and data analysts deployed a trusted research environment using Open OnDemand inside Perth Children's Hospital to facilitate high performance computing on sensitive healthcare data. We developed an extension to Open OnDemand using Podman to enhance security and allow for user apps to securely expose web services to hospital users. This allows modern web applications to be deployed as-is using their standard deployment workflow, typically Dockerfile or Kubernetes based. The deployed extension is in use for multiple applications on healthcare data at Perth Children's Hospital.

Keywords

HPC, containers, security

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1 Introduction

Open OnDemand [4] is a web framework for high performance computing (HPC) users to access and run a variety of customisable HPC applications in a web context, frequently including code-focused applications such as R Studio, Jupyter Lab, VS Code Server, and also SSH and remote desktop access. From a system architecture perspective [5] this involves a coordinating web server namely Apache httpd which communicates and submits jobs to a queuing system via a user-level nginx server running Passenger. Security considerations such as authentication and authorisation are user-configurable via Apache configuration or at the application level in Bash scripts. This leads to significant diversity in approaches to security and often leads to hidden dependencies on host and OS security considerations such as SELinux, network namespaces and filesystem permissions. After customisation, Open OnDemand is a powerful tool to access HPC resources remotely.

The Paediatric Anaesthesia Research Team, part of the Institute for Paediatric Perioperative Excellence (IPPE) at the University of Western Australia, develops and implements clinical research in

perioperative anaesthesia according to consumer priorities [?] including safer anaesthesia for kids, reducing perioperative anxiety and improving care for patients from regional and remote Australia. Our team consists of multidisciplinary researchers including clinicians, software engineers, data analysts and research assistants. In 2024 we received a grant from the Stan Perron Charitable Foundation to deploy a small on-premises HPC system called MERLIN to allow a diverse range of users from less-technical to highly technical to run local artificial intelligence (AI) on sensitive healthcare data. Our design requirements called for a system that was secure, interoperated with internal enterprise authentication and authorisation, allowed for sharing of limited compute resources and was accessible to our less technical users. We chose Open OnDemand as the primary mode of access to MERLIN.

The primary use case of our cluster is to design and deploy AI applications in a web context. The most straightforward way to host these applications is as Open OnDemand applications which will be run in a secure self-contained environment in the user context with access to the relevant data. However, there are a number of challenges to doing this natively in Open OnDemand. Accordingly we designed and built a software service called MERLIN OnDemand Proxy to simplify these bespoke deployments.

2 Prior Work

Trusted research environments have been described by the Australian Research Data Commons (ARDC) as highly secure computing environments that provide remote access to sensitive data for analysis and research [6]. The Australian government recommends the use of the Five Safes approach developed in the United Kingdom when working with sensitive data for research; Safe People, Safe Projects, Safe Data, Safe Settings and Safe Outputs. TREs lie within the Safe Settings part of this framework.

According to the Five Safes framework, TREs should be designed to allow external researchers to access sensitive data that has been deidentified in a secure trusted way. This is the approach taken by other extant TREs in Australia such as QCIF KeyPoint, Secure Unified Research Environment (SURE), UNSW ERICA, Monash Secure Research Platform (SeRP) and Curtin SeRP which provide encrypted access to Windows virtual desktop instances with access to deidentified sensitive datasets. The NHS Secure Data Environment Service also follows this approach by providing a secure Databricks instance for researchers to interact with pseudonymised datasets. This allows researchers to both securely access data within a restricted environment and transfer deidentified data outside of the hospital network environment into an external TRE hosted at a research institution.

Outside of the Five Safes framework, TREs vary considerably in scope and design. One example is Finland's CSC - IT Center for

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Science service ePouta, which facilitates deployments of virtual machines (VMs) and storage resources directly into target institutions using encrypted containerised virtual environment software stacks [2]. Another example is Sweden's Bianca HPC system hosted at Uppsala University, which creates a virtualised cluster with isolated storage and network volumes per project to avoid inter-project contamination [1].

3 Design

The MERLIN cluster was co-designed with our enterprise security team according to the Five Safes framework. We used Red Hat Enterprise Linux as the OS with SELinux enabled, Slurm for job scheduling, rootless Podman as well as Apptainer for container support and Microsoft Entra ID for network authorisation. Due to corporate network restrictions, we host Open OnDemand from one domain and disallow subdomains.

We identified the following common requirements for modern web services.

- (1) An Open Container Initiative (OCI)-compatible container service
- (2) Isolation of internal and external network services such that internal databases and services are not externally accessible
- (3) Authorisation and authentication
- (4) Root domain routing - hosting applications under a URL sub path is frequently not supported

The default Open OnDemand architecture [5] leaves these requirements out of its design and up to user specification. The most commonly suggested solution is to use Open OnDemand's built-in reverse proxy, which routes requests via a sub-path `/rnode/<host>/<port>/<application-specific-path>`, along with an additional authorisation server such as Python's Twisted package. However, this configuration requires the application to support sub-path URL rewriting and leaves open the possibility of brute-forcing ports to gain access to internal services and is therefore not suitable for our use-case. Another possibility may have been to host web services on automatically generated subdomains that are only accessible to users while a job is being run, however due to enterprise network limitations we were unable to access dynamic subdomain generation.

4 Solution

We created a defense-in-depth solution to this problem.

Firstly, we separated internal and external network service boundaries by creating a common Open OnDemand containerised design that enforces a single entry point Caddy web server. We additionally used the OOD Proxy project from Brigham Young University [3] to generate a per-job Linux network namespace to further isolate users from external access. The same approach can be trivially extended to a Kubernetes Pod.

We use the Slurm dependency munge to generate authorisation tokens for individual user jobs. These are stored in the user browser cookies as a compromise between frequent re-authentication and job security. At runtime, the Caddy server verifies that the user supplies a correct and current authorisation token for the given job.

To solve the stateful problem of root domain routing, we created a custom Caddy module that allows Slurm jobs to register their application host and port combinations on the Open OnDemand controller host at job creation time. At runtime, the user is routed to the correct application on the root domain using their authorisation cookie. The Caddy module verifies that user is authenticated, the Slurm job is still running and the validity of the munge token before reverse proxying to the job host.

The final system design is illustrated in Figure 2.

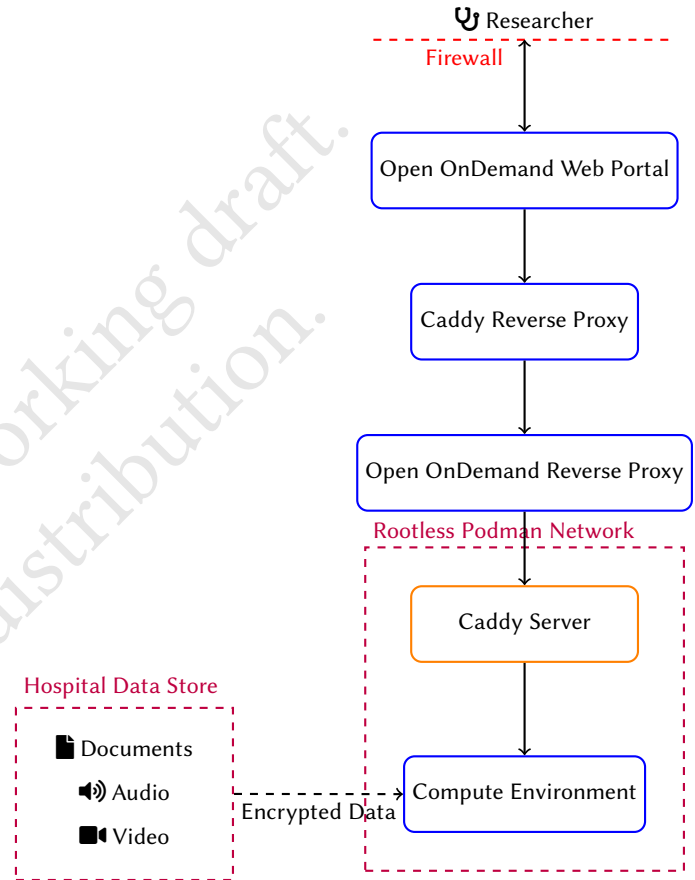


Figure 2: MERLIN OnDemand Proxy workflow

5 Discussion

The presented proxy workflow has enabled easy deployments of web applications in a trusted hospital environment. The Paediatric Anaesthesia Research Team has commenced translational studies including using LLM services like Open WebUI to convert unstructured data from the digital medical record in Perth Children's Hospital into structured data for risk prediction in paediatric anaesthesia. The use of the proxy means that web applications can be introduced by the technical team that are secure, verifiable and self-contained.

The most prominent limitation currently is the maintenance burden and overhead of using a custom authorisation solution. We have endeavoured to keep the design as simple and flexible as possible however it does introduce complexity in deploying

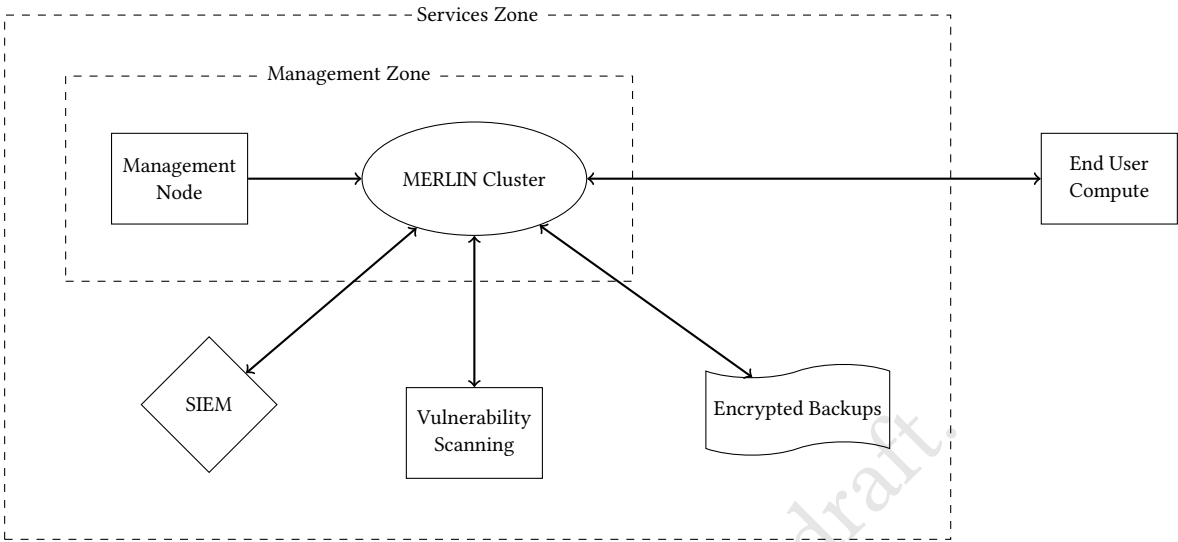


Figure 1: Logical network layout

another service on top of Open OnDemand. Future work could include streamlining this complexity within Open OnDemand.

6 Conclusion

This paper presents the design and deployment of a configurable proxy for Open OnDemand to support modern web applications. The launched environment provides a secure platform for medical research using modern computational techniques such as LLMs, containerisation and defense in depth.

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